AIML- Section 5

ID- 2024AC05653

Name - D. Mallikarjuna Reddy.

(Q1): Data: 4, 5, 7, 8, 9, 10, 11, 13, 14, 15, 100

It's already in sorted order.

total $n = 11$

(i) 5-point Summary

$\boxed{min = 4}$

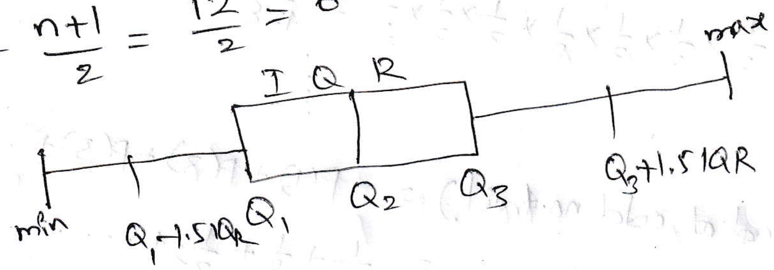
$\boxed{max = 100}$

$\boxed{Q_1} = \left(\frac{n+1}{4}\right)^{th} = \frac{12}{4}^{th} = 3^{rd}$ value $= 7$

$\boxed{Q_3} = 3 \cdot \left(\frac{n+1}{4}\right)^{th} = 9^{th}$ value $= 14$

$\boxed{Q_2} = \frac{n+1}{2} = \frac{12}{2} = 6^{th}$ value $= 10$

Outliers:



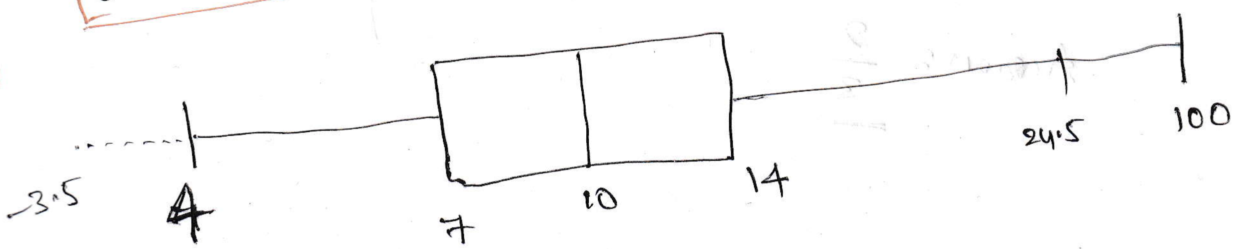
$IQR = Q_3 - Q_1 = 14 - 7 = 7$

Lower Limit $= Q_1 - 1.5 IQR = 7 - 1.5 \times 7 = -3.5$

upper Limit $= Q_3 + 1.5 IQR = 14 + 1.5 \times 7 = 24.5$

$\boxed{\text{Outliers} \Rightarrow 100 \text{ only.}}$

(ii)



Boxplot

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(Q2) Flip a coin until you get the first Head. what is the probability that the no. of flips is odd?

(A) $S = \{H, T\}$ $P(H) = 1/2$; $P(T) = \frac{1}{2}$

first H $\rightarrow$ Stop

flip 1 : $\frac{1}{2} \rightarrow$ stop

else

flip 3: $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} \rightarrow$ stop

else

flip 5: $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{32} \rightarrow$ stop

else

flip 7: $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{128} \rightarrow$ stop

$\vdots$

P(first Head at odd no. trail) $= P(1) + P(3) + P(5) + \ldots$

$= \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \ldots$

$\boxed{\text{Geometric Sum} = \frac{a^{\leftarrow \text{first term}}}{1 - r_{\uparrow \text{ratio}}}}$ $= \frac{\frac{1}{2}}{1 - (\frac{1}{4})} = \frac{1/2}{3/4} = \frac{2}{3}$

for every 2 out 3 times, their is a chance that Head comes.

Answer: $\frac{2}{3}$

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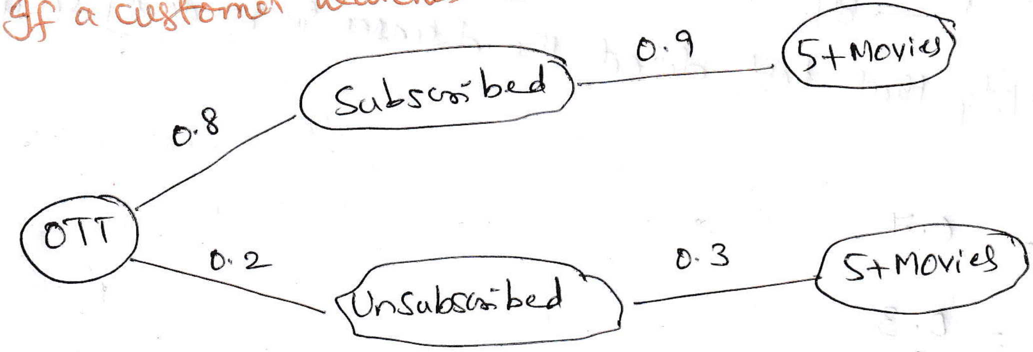
(Q3). 80% remain subscribed; 20% cancelled subscription.

90% watch 5 movies or more   30% watch 5 movies or more

If a customer watches 5 movies or more, probability of him/her subscribed



$$P\left(\frac{Sub}{5+Movie}\right) = \frac{P\left(\frac{5+Movies}{Sub}\right) \cdot P(Sub)}{P(Sub) \cdot P\left(\frac{5+Movies}{Sub}\right) + P(Unsub) \cdot P\left(\frac{5+Movies}{Unsub}\right)}$$

$$= \frac{0.9 \times 0.8}{0.9 \times 0.8 + 0.2 \times 0.3} = \frac{0.72}{0.78} = 0.92307$$

$$\boxed{\approx 92.3\%}$$

* if a customer is watching 5 or more movies per month, then there is high chance (92.3%) that they remain subscribed.

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(Q4) Virus A — 70% cases

Virus B — 30% cases

Test < A — 95%, B — 80%

overall probability that test detect the diseases in randomly selected patient

(A) $P(A) = 0.7$

$P(B) = 0.3$

X – disease is detected

$P(\frac{X}{A}) = 0.95$

$P(\frac{X}{B}) = 0.80$

Probability Disease Detected

$$P(X) = P(A) \cdot P(\tfrac{X}{A}) + P(B) \cdot P(\tfrac{X}{B})$$

$$= 0.7 \times 0.95 + 0.3 \times 0.80$$

$$= 0.665 + 0.24$$

$$= 0.905$$

$$\approx 90.5\%$$

$\boxed{90.5\%}$ chance that the test will detect a patient with disease.